

MA 547: Financial Engineering

Course Code: MA 547

Course Name: Financial Engineering

L-T-P-C: 3-1-0-4

Prerequisites: Probability and Statistics, Calculus.

Intended for: UG, PG, PhD

Mutual Exclusion: NA

Elective or Core: Elective

Approval: 65th BoA

Preamble

Derivatives are financial instruments whose value derives from an underlying asset, and pricing and managing them is a core skill in quantitative finance. This course introduces financial engineering through that lens — how derivative contracts work, how they are priced, and how they are used to trade and to control risk. Intended primarily for B.Tech. students and open as an elective to postgraduate and doctoral students, it assumes no prior finance background and uses only the essential mathematics, so it can be taken as a standalone course. Moving from financial markets and arbitrage-free pricing to forwards, futures and options, trading strategies and hedging, every topic is applied to real Indian and foreign market data, culminating in a project in which students price, trade and hedge a derivative-based strategy.

Course Modules (Total: 42 Lectures)

- **Financial Markets and Derivative Instruments:** Structure of financial markets and their participants, and basic terminology. Forwards, futures and options as instruments, their payoff diagrams and contract mechanics including lot size, expiry, margin and settlement. (6 Lectures)
- **Essential Tools and No-Arbitrage Pricing:** Asset returns and log-returns, volatility, and key return distributions. The concept of arbitrage and no-arbitrage pricing, and risk-neutral valuation introduced intuitively. Brownian motion and geometric Brownian motion as a model for stock prices. (7 Lectures)
- **Pricing of Forwards, Futures and Options:** Cost-of-carry pricing of forwards and futures, effect of dividends and transaction costs, forwards versus futures, and basis risk; hedging with futures. Put–call parity, the binomial model, Cox–Ross–Rubinstein (CRR) model, and the Black–Scholes model with its solution, time-dependency and change of variable. The Greeks and their properties, implied volatility, and the volatility smile. Pricing of real futures and option chains and comparison of model prices with observed market prices. (12 Lectures)
- **Option Trading Strategies:** Bullish, bearish and neutral strategies; spreads, straddles, strangles, butterflies and condors. Construction, payoff and profit-and-loss analysis of each strategy, built and back-checked on real option-chain data. (8 Lectures)
- **Hedging and Risk Management:** General principles and types of hedging; delta hedging, delta-neutral and gamma-neutral portfolios; dynamic hedging and hedging error. A practical introduction to Value at Risk for a derivatives book, with scenario and stress testing. Construction, hedging and rebalancing of a portfolio on market data, feeding the course project. (9 Lectures)



Text Books

1. Hull, J. C., *Options, Futures and Other Derivatives*. Pearson Education, 11th Edition, 2022.
2. Capinski, M., and Zastawniak, T., *Mathematics for Finance: An Introduction to Financial Engineering*. Springer, 2nd Edition, 2011.

References

1. Roman, S., *Introduction to the Mathematics of Finance: From Risk Management to Options Pricing*. Springer, 2004.
2. Shreve, S. E. (2004). *Stochastic Calculus for Finance I: The Binomial Asset Pricing Model*. Springer.
3. Shreve, S. E. (2004). *Stochastic Calculus for Finance II: Continuous-Time Models*. Springer.
4. Benninga, S., *Financial Modeling*. MIT Press, 4th Edition, 2014.
5. Korn, R., and Korn, E., *Option Pricing and Portfolio Optimization*. American Mathematical Society, 2000.